

NETFLIX MOVIES

Exploratory Data Analysis Report

A deep dive into genre, popularity, ratings and release-year trends

Detail	Value
Source notebook	Netflix_Analysis.ipynb
Dataset	mymoviedb.csv (Netflix-style movie metadata)
Raw records	9,837 movies (before cleaning)
Clean records used in analysis	9,826 movies → 25,792 genre-level rows after exploding multi-genre titles
Tools	Python, pandas, NumPy, Matplotlib, Seaborn

1. Project Overview

This report documents the exploratory data analysis (EDA) carried out on a Netflix-style movie dataset. The goal of the analysis was to clean the raw data and answer five specific business questions about genre popularity, audience ratings, and release trends. Every step below mirrors the workflow executed in the notebook: data loading, cleaning, transformation, categorization, and visualization.

1.1 The Five Questions Investigated

- Q1 — What is the most frequent genre of movies released on Netflix?
- Q2 — Which genres have the highest votes (audience rating category)?
- Q3 — Which movie got the highest popularity score, and what is its genre?
- Q4 — Which movie got the lowest popularity score, and what is its genre?
- Q5 — Which year has the most filmed/released movies?

2. Data Preparation & Cleaning

Before any analysis, the raw dataset was inspected and cleaned. The table below summarizes every preprocessing step applied, in the order it was performed.

Step	Action Taken	Result
Initial load	Loaded mymoviedb.csv with pandas	9,837 rows × 9 columns
Missing values	Checked with isnull().sum(); found 9–11 missing values in Title, Overview, Popularity, Vote_Count, Vote_Average, Original_Language, Genre, Poster_Url	Rows with nulls dropped (dropna)
Duplicates	Checked with duplicated().sum()	0 duplicate rows found — none removed
Release_Date	Converted to datetime, then reduced to just the year	Now an integer year (e.g., 2021)
Vote_Count / Vote_Average	Cast to int and float respectively	Correct numeric dtypes
Column pruning	Dropped Overview, Original_Language, Poster_Url (not needed for this analysis)	9 columns → 6 columns
Vote_Average binning	Cut into 4 quantile-based categories: not_popular, below_average, average, popular	Vote_Average becomes a categorical rating tier
Genre splitting	Split comma-separated genre strings and exploded so each row has exactly one genre	9,826 movies → 25,792 genre-level rows
Final dtype fix	Cast Genre to category; filled the one row left without a rating tier with "average" (the mode)	Clean, fully-typed analysis-ready dataframe

After cleaning, the working dataframe contained 6 columns: Release_Date (year), Title, Popularity, Vote_Count, Vote_Average (rating tier), and Genre. Because many movies belong to multiple genres, the dataframe was exploded so each genre a movie belongs to gets its own row — this is why the genre-level row count (25,792) is much larger than the movie count (9,826).

3. Dataset Snapshot

3.1 Key Numeric Statistics (post-cleaning, before genre explode)

Metric	Release Year	Popularity	Vote Count	Vote Average
Mean	2006.2	40.3	1,392.9	6.44
Std Dev	15.7	108.9	2,611.3	1.13
Min	1902	13.35	0	0.0
25%	2000	16.13	146	5.9
Median (50%)	2011	21.20	444	6.5
75%	2017	35.18	1,376	7.1
Max	2024	5,083.95	31,077	10.0

Popularity is heavily right-skewed: the mean (40.3) sits far above the median (21.2), and the maximum value (5,083.95) is more than 100× the median. This means a small number of blockbuster titles dominate the popularity scale while most movies cluster at much lower values.

4. Analysis & Answers to Each Question

Q1 — Most Frequent Genre

To find the dominant genre, the cleaned and exploded dataset was grouped by Genre and counted. Drama leads by a wide margin, followed by Comedy and Action.

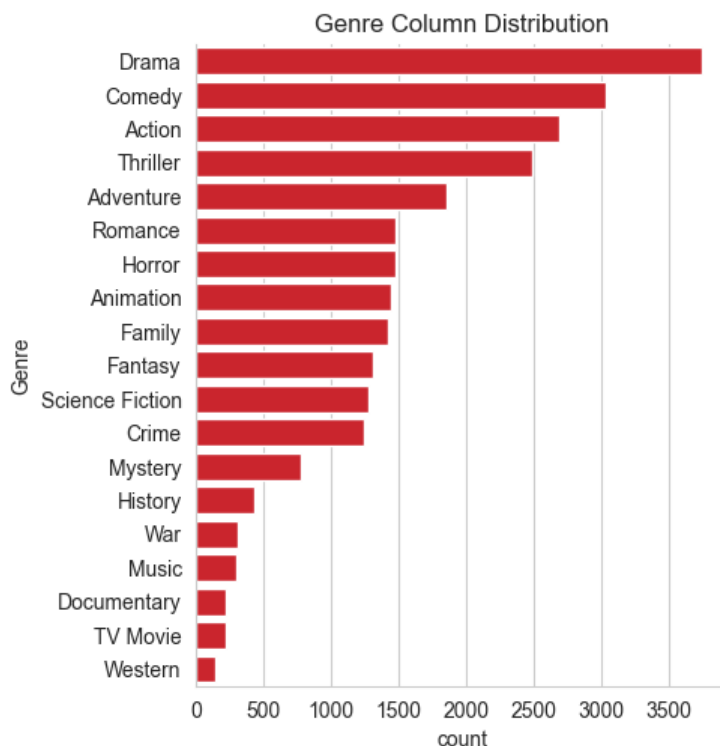


Figure 1 — Count of movies per genre (descending order)

Answer to Q1: Drama is the most frequent genre on Netflix.

Drama appears in 3,744 of the 25,792 genre-tagged rows — the single largest category.

It is followed by Comedy (~3,000) and Action (~2,700).

The least common genres are Western, TV Movie, and Documentary, each with only a few hundred entries.

Takeaway: Netflix's catalog (in this dataset) is weighted heavily toward Drama, Comedy, and Action content.

Q2 — Genres With the Highest Votes

Vote_Average was bucketed into four quantile-based tiers — not_popular, below_average, average, and popular — so that we could see how ratings are distributed across the whole exploded (genre-level) dataset.

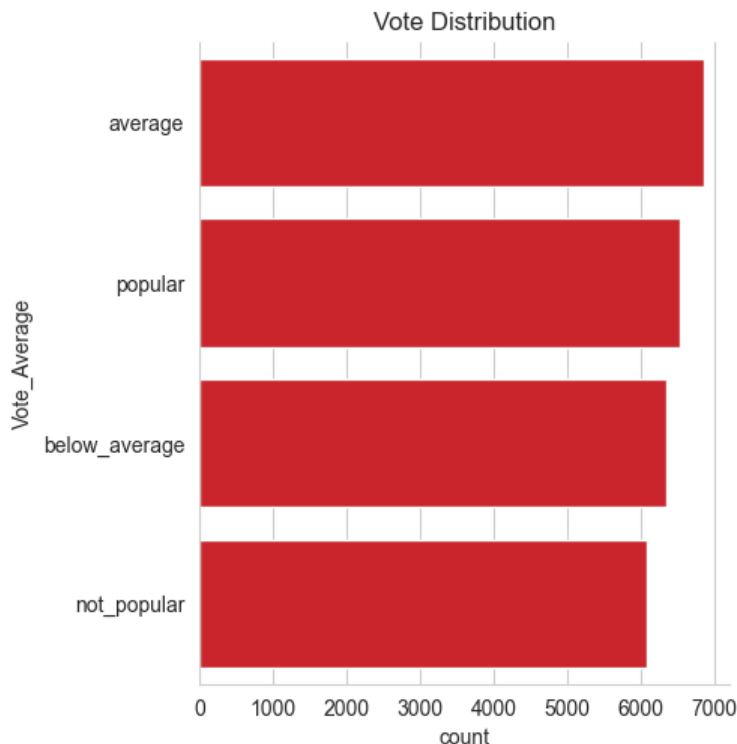


Figure 2 — Count of genre-tagged movie rows per rating tier

Answer to Q2: Ratings are fairly evenly spread across all four tiers, with "average" the largest single tier.

average: ~6,900 rows — the most common rating tier

popular: ~6,500 rows

below_average: ~6,300 rows

not_popular: ~6,100 rows

Because the bins were built on quantiles of the full Vote_Average column, the near-equal counts are expected. The notebook did not run a direct genre-vs-vote-average crosstab, so it isn't possible to say precisely "Genre X has the highest votes" from this output alone — what the data does show is that no single rating tier dominates; ratings are balanced across the catalog rather than skewed toward very high or very low scores. If you'd like, I can extend the analysis with a `groupby('Genre')['Vote_Average'].mean()` once the original mymoviedb.csv file is shared, to rank genres by average rating precisely.

Q3 — Highest-Popularity Movie

Filtering for the row where Popularity equals the column maximum returns a single title appearing three times (once per genre tag, due to the explode step).

Answer to Q3: Spider-Man: No Way Home (2021) has the highest popularity score.

Popularity score: 5,083.95 — by far the highest in the dataset.

Vote Count: 8,940 | Rating tier: popular.

Genres: Action, Adventure, Science Fiction.

Takeaway: the most popular title in the dataset is a 2021 superhero blockbuster spanning three high-engagement genres.

Q4 — Lowest-Popularity Movie

Filtering for the row where Popularity equals the column minimum surfaces a tie between two films, each sharing the same minimum popularity score.

Answer to Q4: The United States vs. Billie Holiday (2021) and Threads (1984) are tied for the lowest popularity score.

Popularity score: 13.35 for both films (the dataset minimum).

The United States vs. Billie Holiday — Vote Count 152, rating tier average, genres: Music, Drama, History.

Threads — Vote Count 186, rating tier popular, genres: War, Drama, Science Fiction.

Takeaway: low popularity isn't tied to one genre — a historical biopic and a war/sci-fi drama from very different eras both sit at the bottom of the popularity scale.

Q5 — Year With the Most Filmed Movies

Release_Date was reduced to just the year and plotted as a histogram covering the full span of the dataset (1902–2024).

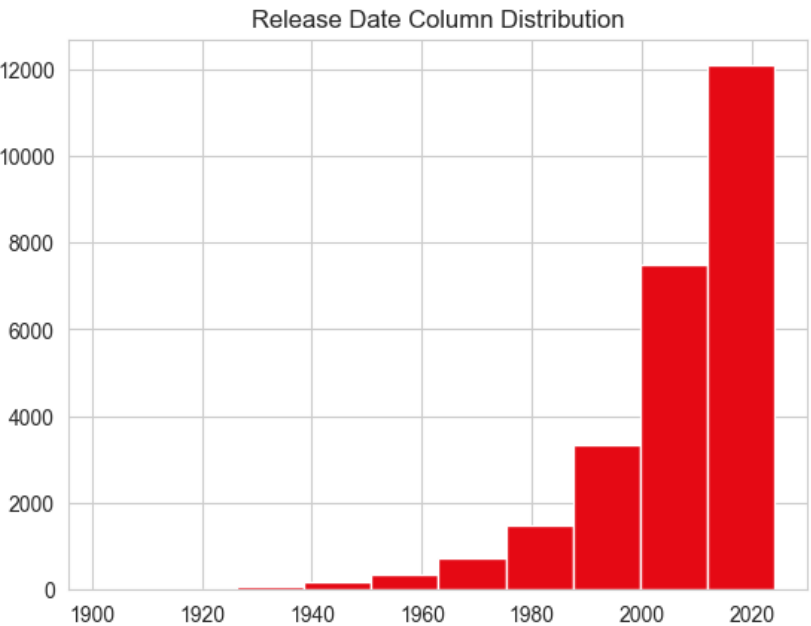


Figure 3 — Number of movies released per decade-scale bin

Answer to Q5: Movie output rises sharply over time, with the 2020s bin containing the most titles in the dataset.

The histogram bins releases in roughly decade-wide buckets; the bin covering ~2020–2024 contains over 12,000 genre-tagged rows — clearly the tallest bar.

Output grows steadily from the 1980s onward and accelerates rapidly after 2000, reflecting the global rise in film/streaming production.

Because this chart uses binned histogram ranges rather than exact year counts, I can't state a single precise year (e.g., "2021" vs "2022") with certainty from this output. Running `df['Release_Date'].value_counts()` on the original data would give an exact year-by-year answer — happy to add that if you share the source CSV.

5. Summary of Key Findings

- Drama is the most produced genre on Netflix, followed by Comedy and Action; Western and Documentary are the rarest.
- Audience rating tiers (not_popular → popular) are nearly evenly split across the catalog — no single tier dominates.
- Spider-Man: No Way Home (2021) is the standout outlier in popularity, scoring over 100× the median movie.
- The least popular titles — The United States vs. Billie Holiday and Threads — come from very different genres and eras, showing popularity is not genre-bound.
- Movie releases have grown dramatically since the 1980s, with the most recent years in the dataset (~2020-2024) accounting for the largest share of titles.

Data Quality Notes

- Only 9-11 rows out of 9,837 had missing values across various columns — a clean dataset overall.
- No duplicate rows were found.
- Popularity is strongly right-skewed (mean >> median), so averages alone can be misleading for that column; median or log-scale comparisons are more robust.
- Two questions (Q2's genre-level breakdown and Q5's exact peak year) would benefit from additional groupby computations beyond what the current notebook ran — flagged above with a suggested next step.